

SOC AUTOMATION PIPELINE PHASE 1

Functional Audit Checklist

Wazuh → Shuffle SOAR → VirusTotal / AbuseIPDB → TheHive → MS Teams

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Purpose: This checklist verifies that each stage of the automated SOC detection-to-response pipeline is functioning as designed, from initial attack detection through enrichment, risk scoring, deduplication, and final case creation/notification.

Instructions: Run a controlled test attack from Kali against the Ubuntu target for each zone, trace the alert through the pipeline, and tick the corresponding boxes below. Record supporting notes and evidence.

ZONE 1 — Detection & Ingestion

The Source — Kali Linux → Ubuntu (Wazuh Agent) → Wazuh Manager → Shuffle SOAR webhook

#	AUDIT CHECK	EXPECTED RESULT	STATUS	NOTES
1	Kali → Ubuntu attack traffic reaches target	SSH brute-force / port scan from Kali is received by Ubuntu VM (SSH/scan traffic visible in logs)	☒ Pass ☐ Fail ☐ N/A	Yes, Logs are visible into the /var/log/auth.log file.
2	Wazuh Agent installed & reporting on Ubuntu VM	Agent status = 'Active' on Wazuh Manager dashboard	☒ Pass ☐ Fail ☐ N/A	Service is running fine.
3	Wazuh Manager rule triggers on attack pattern	Correct rule ID fires (e.g., SSH brute-force rule) with expected severity level	☒ Pass ☐ Fail ☐ N/A	Yes, while multiple password failure then it will automatically generate the Level 10 alert.
4	Wazuh alert forwarded as JSON webhook	Shuffle SOAR webhook endpoint receives well-formed JSON payload with no missing fields	☒ Pass ☐ Fail ☐ N/A	Yes, Shuffle SOAR Webhook receiving multi field JSON Payload from the Wazuh.
5	Shuffle workflow trigger fires reliably	Workflow starts within acceptable latency (define SLA, e.g., <10s) for every alert, no dropped triggers	☒ Pass ☐ Fail ☐ N/A	Workflow Name: Wazuh to Shuffle Automated Alert Processing Status: FINISHED (Successful Execution) Trigger Source: Webhook Execution Duration: ~25 seconds

ZONE 2 — Data Parsing & Triage

The Filter — Parse IOC → Private IP condition → Drop / Proceed

#	AUDIT CHECK	EXPECTED RESULT	STATUS	NOTES
1	Parse IOC (Python) node executes without error	Rule ID, Source IP, Agent Name extracted correctly from JSON webhook payload	☒ Pass ☐ Fail ☐ N/A	It is Extracting the IP Address from the alert.
2	Field extraction accuracy	Extracted values match raw alert JSON exactly (no null/blank fields on valid alerts)	☒ Pass ☐ Fail ☐ N/A	Yes, No Null/Blank Alerts
3	Private IP condition logic	RFC1918 ranges (10.x, 172.16–31.x, 192.168.x) correctly classified as Private	☒ Pass ☐ Fail ☐ N/A	Yes, It is Not parsing Private IP Address further.
4	Drop path for private/internal IPs	Workflow stops cleanly, no downstream nodes execute, no error thrown	☒ Pass ☐ Fail ☐ N/A	If Private IP Found, then it will automatically drop the workflow and completely skip the next actions.
5	Proceed path for public IPs	Workflow correctly routes to Zone 3 (Threat Enrichment) only for public-facing IPs	☒ Pass ☐ Fail ☐ N/A	Yes, If any public IP Found from the Alert, then it will complete the workflow and give the alert in the teams.

ZONE 3 — Threat Intelligence Enrichment

The Brains — GeoIP, VirusTotal, AbuseIPDB → Enrichment Aggregator

#	AUDIT CHECK	EXPECTED RESULT	STATUS	NOTES
1	GeoIP Lookup (ip-api.com) API call	Returns Location, Country, ASN, ISP for valid public IP within timeout window	☒ Pass ☐ Fail ☐ N/A	Yes, It returns City, Country, ASN, Full info from their database.
2	VirusTotal API call	Returns IP Reputation, malicious hit count, engine detections; correct API key / quota not exceeded	☒ Pass ☐ Fail ☐ N/A	Yes, gives the complete report and after that we have parser/enrichment aggregator to get data from the complete report.
3	AbuseIPDB API call	Returns Abuse Confidence Score (0–100%) for the source IP	☒ Pass ☐ Fail ☐ N/A	Yes, It returns with the score of 0-100%
4	Parallel execution behaviour	All three API calls run concurrently, not sequentially blocking one another	☒ Pass ☐ Fail ☐ N/A	Yes, they all are running concurrently not blocking any other.
5	API timeout / failure handling	If one API is down or rate-limited, workflow degrades gracefully (retry/timeout) instead of crashing	☒ Pass ☐ Fail ☐ N/A	If API Doesn't get the information, then it will completely run without them
6	Enrichment Aggregator merge logic	GeoIP + VT + AbuseIPDB results merge into a single, correctly-keyed JSON payload with no data loss	☒ Pass ☐ Fail ☐ N/A	Enrichment working fine and it collect data from all 3 Threat intelligence platform and give the summary.

ZONE 4 — Risk Scoring & Deduplication

The Logic — Risk Score → Fingerprint → TheHive Duplicate Check

#	AUDIT CHECK	EXPECTED RESULT	STATUS	NOTES
1	Risk Scoring Node calculation	Score (out of 120) correctly derived from VT hits + AbuseIPDB score + Wazuh rule level, using documented formula	☒ Pass ☐ Fail ☐ N/A	Yes, giving the calculation based on the documented formula.
2	Severity band assignment (1–4)	Score maps to correct severity colour/level consistently across test cases	☒ Pass ☐ Fail ☐ N/A	Yes, as per calculation level it changes the color/level of the case.
3	Fingerprint generation (MD5)	Hash = MD5(Rule ID + Source IP) is deterministic — same inputs always produce the same fingerprint	☒ Pass ☐ Fail ☐ N/A	Yes, it is creating based on the Rule ID + Source IP.
4	Duplicate check query to TheHive	TheHive is queried by fingerprint tag and returns correct match/no-match result	☒ Pass ☐ Fail ☐ N/A	Yes, there is another node that will check the existing case or duplicate case in the documented time.
5	Duplicate found → STOP path	Workflow halts, no second case created, no duplicate alert noise	☒ Pass ☐ Fail ☐ N/A	Yes, If duplicate issue found then it will stop the Execution.
6	Unique fingerprint → PROCEED path	Workflow correctly continues to Zone 5 (Case Creation) only for genuinely new incidents	☒ Pass ☐ Fail ☐ N/A	Yes, If Unique Case Fingerprint found then it will continue to create case.

ZONE 5 — Case Management & Notification

The Action — TheHive Case → Observable → Task → MS Teams Alert

#	AUDIT CHECK	EXPECTED RESULT	STATUS	NOTES
1	'Promote to Case' creates new TheHive case	New case appears in TheHive with correct title, source, and timestamp	☒ Pass ☐ Fail ☐ N/A	Yes, it creates based on provided details by previous nodes.
2	Markdown description populated	Case description includes Risk Score, GeoIP, and Threat Intel details, correctly formatted (no broken Markdown)	☒ Pass ☐ Fail ☐ N/A	Yes, this things are correctly visible and correctly formatted.
3	Observable (Source IP) attached	Source IP is added as an Observable object on the case, tagged appropriately (e.g., malicious/suspicious)	☒ Pass ☐ Fail ☐ N/A	If The IP Found in the alert, then this node automatically update the observable to previous created case.
4	Investigation task created	Standard task template is auto-added to the case for the analyst to work	☒ Pass ☐ Fail ☐ N/A	Yes, it automatically creates the task.
5	MS Teams webhook notification	Alert card delivered to correct Teams channel with severity, enriched data, and direct case link (link opens the right case)	☒ Pass ☐ Fail ☐ N/A	Yes, MS Teams alert webhook notification is working fine as per documented details.
6	End-to-end case data integrity	All fields in TheHive case match the enriched JSON payload — no missing/corrupted values	☒ Pass ☐ Fail ☐ N/A	Yes, there are no missing/corrupted values.

ZONE E2E — End-to-End & Reliability Checks

Cross-cutting checks spanning the full pipeline (Zones 1–5)

#	AUDIT CHECK	EXPECTED RESULT	STATUS	NOTES
1	Full pipeline dry-run (Kali attack → Teams alert)	Single simulated attack produces exactly one Teams notification and one TheHive case, within target SLA (e.g., < 2 min)	☒ Pass ☐ Fail ☐ N/A	Yes, complete workbook runs within the SLA Time.
2	Duplicate alert suppression under load	Repeated identical attacks within test window generate only 1 case, not N cases	☒ Pass ☐ Fail ☐ N/A	Yes, If Duplicate alert found within time it will not crates the case within time.
3	False-positive / private-IP suppression	Internal-only traffic never reaches Zone 3–5; no noise generated	☒ Pass ☐ Fail ☐ N/A	Yes, After the IOC there is condition for the next node is that if Private IP=True then it will not go further.
4	Credential & API key validity	Wazuh, VirusTotal, AbuseIPDB, TheHive, Cortex, Shuffle, and Teams webhook credentials are current and not near expiry	☒ Pass ☐ Fail ☐ N/A	In this I am using Trail version so it will be expired in few days. This is for Just a Testing purpose.
5	Error/exception logging	Any node failure (API down, malformed JSON, workflow error) is logged and does not fail silently	☒ Pass ☐ Fail ☐ N/A	All nodes are working fine.
6	Workflow performance under concurrent alerts	No race conditions / polling failures when multiple alerts fire close together (known prior issue area)	☒ Pass ☐ Fail ☐ N/A	

AUDIT SUMMARY

ZONE	TOTAL CHECKS	PASS	FAIL	N/A	PENDING
Zone 1 — Detection & Ingestion	5	5	0	0	0
Zone 2 — Data Parsing & Triage	5	5	0	0	0
Zone 3 — Threat Intelligence Enrichment	6	6	0	0	0
Zone 4 — Risk Scoring & Deduplication	6	6	0	0	0
Zone 5 — Case Management & Notification	6	6	0	0	0
Zone E2E — End-to-End & Reliability Checks	6	6	0	0	0

OVERALL PIPELINE STATUS

☒ Fully Operational ☐ Operational with Issues ☐ Not Operational

KEY ISSUES / REMEDIATION ACTIONS

No key issues recorded.

AUDITOR SIGNATURE

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DATE

01/08/2026